# MAST32001 Computational Statistics — Course Assignment

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## Task 2: Bayesian mixture modelling with MCMC

### Part A: Latent-variable representation and the Gibbs sampler

#### Model specification and priors

A two-component mixture describes observations that cluster around two different locations. What we observe is the measurement, not its component label. Introducing the missing label turns one strange likelihood into familiar normal and binomial pieces. For independent observations $y_1, \dots, y_n$, let $Z_i \in \{1, 2\}$ be the latent allocation, with $\Pr(Z_i = 1) = \pi$ and $\Pr(Z_i = 2) = 1 - \pi$. Conditional on its allocation,

$$Y_i \mid Z_i = k, \mu_1, \mu_2, \sigma^2 \sim \mathcal{N}(\mu_k, \sigma^2), \qquad k = 1, 2.$$

Integrating out $Z_i$ gives the familiar mixture density:

$$p(y_i \mid \mu_1, \mu_2, \pi, \sigma^2) = \pi\varphi(y_i; \mu_1, \sigma^2) + (1 - \pi)\varphi(y_i; \mu_2, \sigma^2),$$

This is a two-component normal mixture with a shared variance. The observations are independent given the mixutre parameters.

We use the independent, proper priors

$$\mu_1, \mu_2 \overset{\text{iid}}{\sim} \mathcal{N}(m_0, s_0^2), \qquad \pi \sim \text{Beta}(a, b), \qquad \sigma^2 \sim \text{InvGamma}(\alpha_0, \beta_0),$$

with $m_0 = 0, s_0 = 10, a = b = 1,$ and $\alpha_0 = 2, \beta = 1$. The weight prior is `Beta(1,1)`, which is the uniform prior. The mean prior is broad relative to these data, but it remains important when a component contains few observations. The two mean priors are indentical and the weight prior is symmetric, so the model treats the two labels symmetrically. The variance prior has mean 1 and infinite variance. Thse choice keep every update conjugate.

Our inverse-gamma convention is

$$p(v) \propto v^{-\alpha_0 - 1} \exp(-\beta_0 / v), \qquad v > 0,$$

so $1/v$ is gamma with shape $\alpha_0$ and rate $\beta_0$. This convention fixed the rate used in the `R` sampler.

To derive the updates, let $n_k = \sum_i I(Z_i = k)$ and $S_k = \sum_{i:Z_i=k} y_i$. Given the allocations, each observation contributes to only one normal component. The likelihood is

$$L(\theta, z; y) \propto \pi^{n_1}(1 - \pi)^{n_2}(\sigma^2)^{-n/2} exp\left[-\frac{1}{2\sigma^2}\sum_i (y_i - \mu_{z_i})^2\right].$$

Multiplying by the independent priors gives $p(\theta, z \mid y) \propto L(\theta, z; y)p(\mu_1)p(\mu_2)p(\pi)p(\sigma^2)$. This is the basis of the four conditional updates. Each update keeps only the terms involving its own unknown.

### Full conditional distribution

The complete-data form lets us update one block at a time. Each distribution below conditions on the current values of the other unknown.

**Allocations.** Given the current parameters, Bayes' rule gives the probability that observation $i$ belongs to component 1:

$$\Pr(Z_i = 1 \mid y_i, \theta) = \frac{\pi\varphi(y_i; \mu_1, \sigma^2)}{\pi\varphi(y_i; \mu_1, \sigma^2) + (1-\pi)\varphi(y_i; \mu_2, \sigma^2)} = p_i$$

We sample $I(Z_i = 1) \sim \text{Bernoulli}(p_i)$ independently across observations. To avoid numerical underflow, it first calculates

$$\ell_{i1} = \log \pi + \log \varphi(y_i; \mu_1, \sigma^2), \quad \ell_{i2} = \log(1-\pi) + \log \varphi(y_i; \mu_2, \sigma^2),$$

then uses $p_i = \text{logit}^{-1}(\ell_{i1} - \ell_{i2})$.

**Component means.** Once observations have been allocated, each component has a normal likelihood. For $k = 1, 2$, combing it with the mean prior gives

$$p(\mu_k \mid -) \propto exp\left[-\frac{(\mu_k - m_0)^2}{2s_0^2} - \frac{1}{2\sigma^2}\sum_{i:Z_i=k}(y_i - \mu_k)^2\right].$$

Completing the square gives

$$V_k = \left(s_0^{-2} + \frac{n_k}{\sigma^2}\right)^{-1}, \quad M_k = V_k\left(\frac{m_0}{s_0^2} + \frac{S_k}{\sigma^2}\right), \quad \mu_k \mid - \sim \mathcal{N}(M_k, V_k).$$

If $n_k = 0$, then $S_k = 0$ and the update reduces to the prior. We will use this fact later, because it explains the large mean excursions in the overlap run.

**Mixing proportion (Weight).** The sampled counts contribute $\pi^{n_1}(1-\pi)^{n_2}$ to the likelihood. Combining this with the beta prior gives

$$\pi \mid - \sim \text{Beta}(a + n_1, b + n_2).$$

**Variance.** Finally, pool the residuals from both components as $\text{SSE} = \sum_i (y_i - \mu_{Z_i})^2$. Combining likelihood and prior gives

$$p(\sigma^2 \mid -) \propto (\sigma^2)^{-(\alpha_0 + n/2)-1} exp\left[-\frac{\beta_0 + \text{SSE}/2}{\sigma^2}\right],$$

and hence

$$\sigma^2 \mid - \sim \text{InvGamma}(\alpha_0 + \frac{n}{2}, \beta_0 + \frac{\text{SSE}}{2}).$$

The mean prior do not involve $\sigma^2$, so they add no extra term to this conditional shape.

### R implementation

Our own `rgibbs_mix()` in `R/task2.R` updates labels, both means, mixing proportion and variance in that order, using the latest values. This excerpt shows numerical steps from the full function:

```
# 1. Allocations
log1 <- log(pi_c) + dnorm(y, mu1_c, sqrt(s2_c), log = TRUE)
log2 <- log1p(-pi_c) + dnorm(y, mu2_c, sqrt(s2_c), log = TRUE)
p1 <- plogis(log1 - log2); zi <- rbinom(n, 1, p1)
n1 <- sum(zi == 1); n2 <- n - n1
# 2. Component means
ybar1 <- if (n1 > 0) mean(y[zi == 1]) else 0
ybar2 <- if (n2 > 0) mean(y[zi == 0]) else 0
```

```r
prec1 <- 1 / s0^2 + n1 / s2_c; prec2 <- 1 / s0^2 + n2 / s2_c
mu1_c <- rnorm(1, (m0 / s0^2 + n1 * ybar1 / s2_c) / prec1, sqrt(1 / prec1))
mu2_c <- rnorm(1, (m0 / s0^2 + n2 * ybar2 / s2_c) / prec2, sqrt(1 / prec2))
# 3. Mixing proportion
pi_c <- rbeta(1, a + n1, b + n2)
# 4. Variance: mean priors are independent N(m0,s0^2). I
resid <- sum(ifelse(zi == 1, (y - mu1_c)^2, (y - mu2_c)^2))
s2_c <- 1 / rgamma(1, shape = alpha0 + n / 2, rate = beta0 + resid / 2)
```

Log densities prevent numerical underflow. The reciprocal-gamma step uses our chosen rate and residuals from the newly drawn means. The function checks that inputs and states are finite, the variance is positive and the mixing proportion lies strictly between 0 and 1.

We save all parameter draws and $n_1$, keeping both the full paths and the draws after burn-in. These give the posterior summaries, traces, switch counts and allocation probabilities. Two fixed start differ in mean positions, labels, mixing proportions and variances. We give overlap more iterations because it mixes slowly, then use traces, chain agreement, $\hat{R}$ and ESS to assess the run lengths in `Part B`.

## Part B: Simulation study and computational behavior

### Simulation design

We first generate data from a known mixture, so we can compare the posterior with the values that produced the sample. Each setting has $n = 500$, $\pi = 0.5$, and $\sigma^2 = 1$, The separated has means $(-2, 2)$ and the overlap setting has means $(-0.5, 0.5)$.

We use data seed and the same latent labels in both settings, with new normal draws for the observations. Keeping the sample size, weights, variances and priors the same help us focus on the effect of overlap. There is one sample per setting, so this checks the sampler on these examples rather than measuring bias or coverage across repeated samples.

Each dataset is fitted from two different starts. The `swap` stat puts the means in the reverse order and the `spread` start places them farther apart. Both simulated settings use the same starts in Table 1. These are the sampler's starting guesses. They do not change the true means used generated the data. The Faithful rows are included for the later analysis in `Part D`.

**Table 1.** Initial values and seeds for the two chains in each setting. The last two rows are the Old Faithful chains analysed in Part D.

| Setting | Chain | Seed | mu1 | mu2 | pi | s2 |
|---|---|---|---|---|---|---|
| Separated | swap | 101 | 2.000 | -2.000 | 0.9 | 1.000 |
| Separated | spread | 102 | -4.000 | 4.000 | 0.5 | 4.000 |
| Overlap | swap | 201 | 2.000 | -2.000 | 0.9 | 1.000 |
| Overlap | spread | 202 | -4.000 | 4.000 | 0.5 | 4.000 |
| Faithful | swap | 301 | 4.533 | 2.003 | 0.8 | 1.303 |
| Faithful | spread | 302 | 0.459 | 6.241 | 0.5 | 2.605 |

```r
# Two settings: clearly separated vs substantially overlapping components
n <- 500; set.seed(20261016); z_true <- rbinom(n, 1, 0.5)
y_sep <- ifelse(z_true == 1, rnorm(n, -2, 1), rnorm(n, 2, 1))      # mu = +/- 2
y_ov  <- ifelse(z_true == 1, rnorm(n, -0.5, 1), rnorm(n, 0.5, 1))  # mu = +/- 0.5

# Two chains from deliberately different starting values
inits <- list(
  swap   = list(mu1 = 2, mu2 = -2, pi = 0.9, s2 = 1),  # swapped labels
  spread = list(mu1 = -4, mu2 = 4, pi = 0.5, s2 = 4)    # spread means
)
```

```
run_chains <- function(y, inits, n_iter = 5000, burn = 1000, seed_base = 100) {
  ans <- lapply(seq_along(inits), function(i) {
    message("Chain ", i, ": ", names(inits)[i])
    rgibbs_mix(y, n_iter = n_iter, burn = burn, init = inits[[i]], seed = seed_base + i)
  })
  setNames(ans, names(inits))
}
message("Separated simulation")
chains_sep <- run_chains(y_sep, inits, n_iter = 20000, burn = 5000, seed_base = 100)
message("Overlapping simulation")
chains_ov  <- run_chains(y_ov, inits, n_iter = 60000, burn = 15000, seed_base = 200)
```

### Recovery of the data-generating mixture

To compare draws with the truth, we first name the lower-mean component consistently. Whenever $\mu_{(1)} < \mu_{(2)}$, we exchange the means and replace $\pi$ with $1 - \pi$. The common variance stays unchanged. Table 2 combines both chains after this transformation.

```
# Relabel by ordering the means: mu(1) < mu(2); flip pi accordingly
relabel_chain <- function(ch) {
  s <- ch$samples; flip <- s$mu1 > s$mu2
  data.frame(mu_lo = pmin(s$mu1, s$mu2), mu_hi = pmax(s$mu1, s$mu2),
             pi_lo = ifelse(flip, 1 - s$pi, s$pi), s2 = s$s2,
             n_lo = ifelse(flip, ch$n - s$n1, s$n1))
}
pool_relabeled <- function(chains) do.call(rbind, lapply(chains, function(ch)
  relabel_chain(ch)[, c("mu_lo", "mu_hi", "pi_lo", "s2")]))
```

**Table 2.** Parameter recovery: posterior medians and central 95% credible intervals, using all retained ordered draws.

| Setting | Parameter | Truth | Median | 95% interval |
|---|---|---|---|---|
| Separated | mu1 (lower) | -2.0 | -1.98 | [-2.10, -1.86] |
| Separated | mu2 (upper) | 2.0 | 2.04 | [1.91, 2.18] |
| Separated | pi | 0.5 | 0.54 | [0.49, 0.58] |
| Separated | sigma2 | 1.0 | 0.91 | [0.80, 1.05] |
| Overlap | mu1 (lower) | -0.5 | -0.40 | [-2.58, -0.12] |
| Overlap | mu2 (upper) | 0.5 | 0.08 | [-0.20, 2.25] |
| Overlap | pi | 0.5 | 0.49 | [0.01, 0.99] |
| Overlap | sigma2 | 1.0 | 1.14 | [0.88, 1.34] |

The separated estimates are close to their generating values. Under overlap, the means and weight are much less precisely identified: the upper-mean median is 0.08, compared with the true value 0.5, and the weight interval spans [0.01, 0.99]. Interval coverage in this one sample does not establish accurate parameter recovery. The main finding is that two weakly separated components can describe the data without having individually well-determined parameters.

### Trace plots and convergence diagnostics

The raw means in Figure 1 tell us why starting values and labels need attention. Under separation the chains stay in opposite labels. Under overlap, the ordering changes often and the means sometimes travel far from the data. An almost empty component can also produce a large excursion, so a large jump alone does not identify a label switch. `Part C` examines this more clearly.

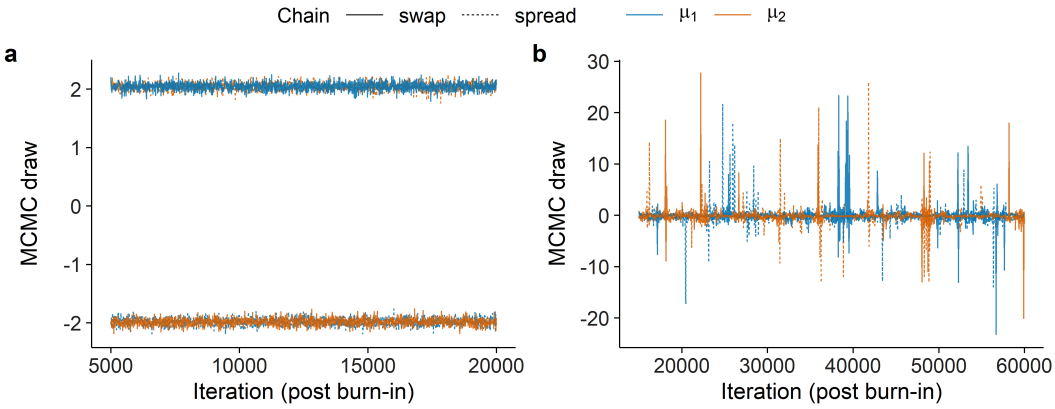



**Figure 1.** Raw mean traces for teh separated (a) and overlap (b). Colors distinguish the means; solid and dashed lines distinguish the two starts.Overlap shows label changes and large excursions when a component is nearly empty.

We then compare the chains after relabeling, so their component names agree. Rank-normalized split $\hat{R}$ checks agreement across chains, while bulk and tail ESS describe how much information remains after accounting for dependence between draws [1]. We use $\hat{R} \leq 1.01$ and both ESS measures at least 400 for every ordered parameter as a practical screen.

**Table 3.** Computational behaviour across settings. Run lengths are per chain, diagnostics combine both ordered chains, and sampler time is summed over the two chains.

| Metric | Separated | Overlap | Faithful |
|---|---|---|---|
| Iterations per chain | 20000 | 60000 | 20000 |
| Burn-in per chain | 5000 | 15000 | 5000 |
| Retained draws per chain | 15000 | 45000 | 15000 |
| Maximum ordered Rhat | 1.0001 | 1.0067 | 1.0001 |
| Minimum bulk ESS | 20826 | 2878 | 27119 |
| Minimum tail ESS | 25182 | 1756 | 28254 |
| Sampler time (seconds) | 4.70 | 14.28 | 2.78 |
| Ambiguous allocations (%) | 2.00 | 100.00 | 0.37 |

Both simulated setting pass this screen. Even with the longer run, overlap gives fewer effective draws and takes more sampler time. The extra iterations help control Monte Carlo error, but they do not remove uncertainty about the mixture itself. Passing the screen supports these summaries. It does not guarantee that every posterior region has been explored. Full parameter-level diagnostics and MCSEs remain in the saved CSV files. Recorded times depends on the machine used.

### Agreement across starts

Table 4 compares the starts. Their ordered medians are close. Switches count changes in the sign of $\mu_1 - \mu_2$ between successive retained draws.

**Table 4.** Per-chain posterior medians after relabeling (order restriction $\mu_{(1)} < \mu_{(2)}$) and label-switch counts.

| Setting | Chain | mu_lo median | mu_hi median | pi_lo median | Switches |
|---|---|---|---|---|---|
| Separated | swap | -1.979 | 2.044 | 0.537 | 0 |
| Separated | spread | -1.982 | 2.044 | 0.536 | 0 |
| Overlap | swap | -0.411 | 0.082 | 0.475 | 2984 |
| Overlap | spread | -0.395 | 0.073 | 0.498 | 3442 |
| Faithful | swap | 2.048 | 4.297 | 0.361 | 0 |
| Faithful | spread | 2.049 | 4.297 | 0.361 | 0 |

### Fitted densities and allocation uncertainty

Next, Figure 2 averages the fitted mixture density over posterior draws. The overall fit stays close to the generating curve, even when individual component parameters are uncertain.

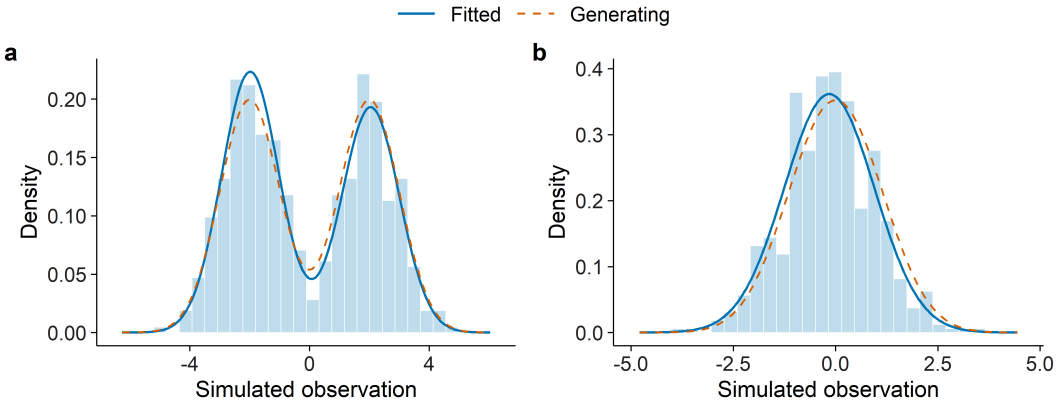



**Figure 2.** Simulated data with the fitted posterior predictive density (solid) and the true generating density (dashed): separated (a) and overlap (b).

Figure 3 averages each observation's conditional probability of belonging to the lower-mean component over both chain. Values near 0 or 1 show clear allocations. We call $0.2 < p_i < 0.8$ ambiguous. These descriptive cutoffs define the fraction in Table 3.

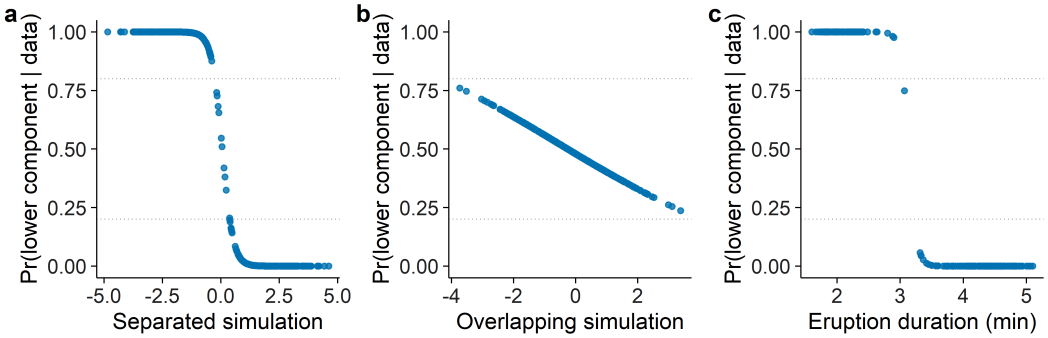



**Figure 3.** Lower-component allocation probabilities: separated (a), overlap (b), Faithful (c).

Separation gives clear parameter and memberships. With overlap, a close overall fit leaves both uncertain. `Part C` examines how label switching affects the component summaries.

## Part C: Label switching and identifiability

### The symmetry of the likelihood

Part B showed label switching happening in practice. Here we look at why it happens and what can be done about it. The reason is a symmetry of the model. The mixture density is invariant under permutation of the component labels:

$$p(y_i \mid \mu_1, \mu_2, \sigma^2, \pi) = p(y_i \mid \mu_2, \mu_1, \sigma^2, 1 - \pi),$$

The same invariance holds for the posterior, because the prior is symmetric in $\mu_1$ and $\mu_2$. The parameter vector $(\mu_1, \mu_2, \pi)$ is then identifiable only up to this permutation. The posterior has two equivalent modes. A chain that can cross the low-probability region between them alternates between the two. Label switching is a feature of the model itself, not an artefact of our implementation.

### Identifiability restrictions and post-processing

Two remedies are available. The first is a constrained prior, for example $\mu_1 < \mu_2$ a priori. It makes the model identifiable by construction, but it changes the target posterior and complicates the sampler. The second, which we use here, is to run the sampler unconstrained and relabel the draws afterwards. We use the order restriction $\mu_{(1)} < \mu_{(2)}$. At each iteration we assign the smaller and the larger of $(\mu_1, \mu_2)$ to the two components and flip $\pi$ accordingly ($\pi \mapsto 1 - \pi$ when $\mu_1 > \mu_2$).

Here the two mean priors and the Beta$(1, 1)$ weight prior are symmetric. With the same priors and proper normalisation, imposing $\mu_1 < \mu_2$ gives the same ordered posterior as sorting unconstrained draws. It fixes the component names, but does not make overlapping components easier to distinguish [2], [3].

**Table 5.** Component-mean posterior medians before and after relabelling, by chain. All retained draws are used.

| Setting | Chain | Raw mu1 | Raw mu2 | Ordered low | Ordered high |
|---|---|---|---|---|---|
| Separated | swap | 2.04 | -1.98 | -1.98 | 2.04 |
| Separated | spread | -1.98 | 2.04 | -1.98 | 2.04 |
| Overlap | swap | -0.14 | -0.17 | -0.41 | 0.08 |
| Overlap | spread | -0.17 | -0.14 | -0.39 | 0.07 |
| Faithful | swap | 4.30 | 2.05 | 2.05 | 4.30 |
| Faithful | spread | 2.05 | 4.30 | 2.05 | 4.30 |

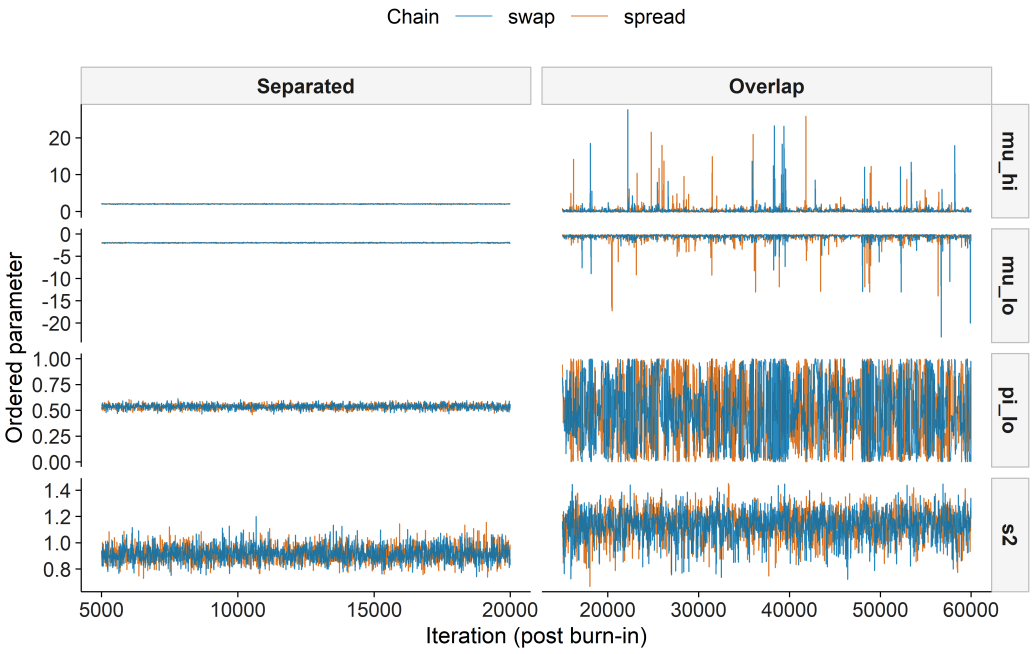



**Figure 4.** Order-restricted traces (two chains, one colour per chain) for the separated and the overlapping settings. After relabeling by $\mu_{(1)} < \mu_{(2)}$, the switching disappears and the chains converge to a common, interpretable parameterization; the overlapping chains still mix more slowly than the separated ones.

### Critique of simple order restrictions ($\mu_{(1)} < \mu_{(2)}$)

While post-processing with $\mu_{\text{lo}} = \min(\mu_1, \mu_2)$ and $\mu_{\text{hi}} = \max(\mu_1, \mu_2)$ resolves label assignment, it introduces two distinct artifacts under heavy component overlap:

1. **Order-statistic truncation (repulsion bias):** Because marginal posterior distributions overlap across zero, folding the parameter space across $\mu_1 = \mu_2$ truncates the tails, artificially repelling posterior medians away from each other ($E[\min(\mu_1, \mu_2)] \leq \min(E[\mu_1], E[\mu_2])$).
2. **Prior tails from empty components.** In % of overlapping sweeps, an empty component ($n_k = 0$) causes $\mu_k$ to update directly from the diffuse prior $N(0, 10^2)$. The order statistic systematically channels extreme negative prior draws into $\mu_{\text{lo}}$ and positive draws into $\mu_{\text{hi}}$, generating skewed credible intervals

([-2.58, -0.12] and [-0.20, 2.25]). Conditioning on non-empty occupancy ($\min(n_1, n_2) \geq 5$) successfully eliminates these prior-tail excursions.

The inequality above concerns expectations, so it does not by itself establish bias in posterior medians. Ordering defines the lower and upper component means. The occupancy threshold removes empty and very small components, but does not guarantee that every extreme draw disappears. It also changes the target to $p(\theta \mid y, \min(n_1, n_2) \geq 5)$, so we use it as a sensitivity analysis and keep all ordered draws for the main results.

The joint draws in Figure 5 show the empty-component excursions. Only the plotted points are thinned.

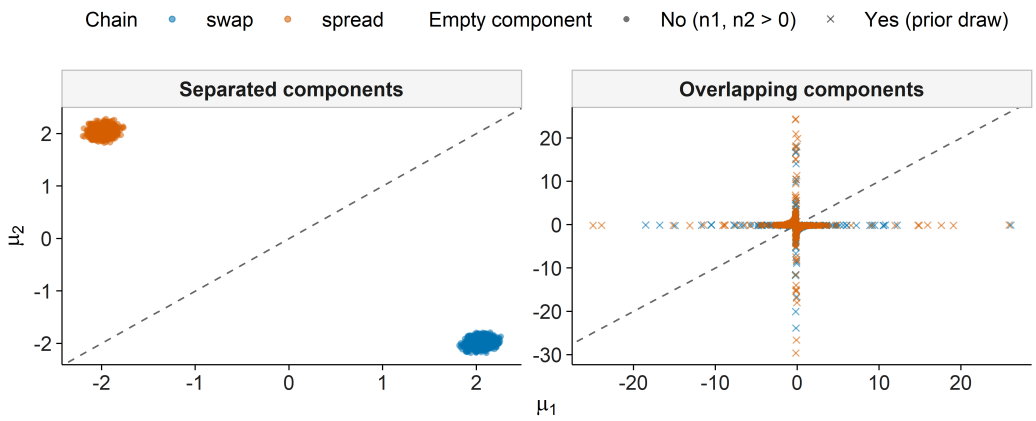



**Figure 5.** Joint raw mean draws for separated and overlapping components. Crosses identify sweeps with an empty component; each panel uses its own axis scales.

**Table 6.** Fraction of retained sweeps with an empty component or an extreme mixing weight.

| Setting | Empty-component draws (%) | Extreme weight draws (%) |
|---|---|---|
| Separated | 0.00 | 0.00 |
| Overlap | 3.56 | 7.33 |
| Faithful | 0.00 | 0.00 |

The current overlap run retains 92.57% of draws under the occupancy rule and removes 7.43%. Table 7 shows how the summaries change.

We use the following filter to check how small component counts affect the posterior summaries.

```
ord_all <- do.call(rbind, lapply(chains_ov, relabel_chain))
n_obs <- chains_ov[[1]]$n
min_occ <- 5
keep <- ord_all$n_lo >= min_occ &
  (n_obs - ord_all$n_lo) >= min_occ
ord_sub <- ord_all[keep, , drop = FALSE]
```

**Table 7.** Overlap: all ordered draws and occupancy-conditioned draws.

| Scheme | Parameter | Retained (%) | Median | 95% interval |
|---|---|---|---|---|
| All | mu_lo | 100.00 | -0.40 | [-2.58, -0.12] |
| All | mu_hi | 100.00 | 0.08 | [-0.20, 2.25] |
| All | pi_lo | 100.00 | 0.49 | [0.01, 0.99] |
| All | s2 | 100.00 | 1.14 | [0.88, 1.34] |
| Conditional | mu_lo | 92.57 | -0.40 | [-1.73, -0.13] |
| Conditional | mu_hi | 92.57 | 0.08 | [-0.19, 1.37] |
| Conditional | pi_lo | 92.57 | 0.49 | [0.02, 0.98] |
| Conditional | s2 | 92.57 | 1.14 | [0.88, 1.33] |

### Additional label-invariant summaries

The predictive density, common variance, mixture mean $m = \pi\mu_1 + (1-\pi)\mu_2$, mixture variance $v = \sigma^2 + \pi(1-\pi)(\mu_1-\mu_2)^2$ and separation $|\mu_1 - \mu_2|$ do not change when labels are exchanged. Table 8 uses the raw common-variance draws. A probability of belonging to the raw component 1 does change with labels; ordering or a label-invariant clustering summary is needed for allocation inference.

**Table 8.** Label-invariant common-variance posterior summaries. Separation is the absolute difference between component means.

| Setting | Quantity | Median | 95% interval |
|---|---|---|---|
| Separated | Mixture mean | -0.11 | [-0.31, 0.08] |
| Separated | Mixture variance | 4.93 | [4.58, 5.31] |
| Separated | Mean separation | 4.02 | [3.85, 4.20] |
| Overlap | Mixture mean | -0.16 | [-0.25, -0.06] |
| Overlap | Mixture variance | 1.22 | [1.08, 1.40] |
| Overlap | Mean separation | 0.74 | [0.04, 4.37] |
| Faithful | Mixture mean | 3.49 | [3.35, 3.62] |
| Faithful | Mixture variance | 1.30 | [1.18, 1.43] |
| Faithful | Mean separation | 2.25 | [2.16, 2.34] |

## Part D: Application to the Old Faithful geyser data

So far everything has been simulated. To see whether the sampler works in practice, we fitted the two-component mixture to the eruption durations of the Old Faithful geyser ($n = 272$, from the `datasets` package) [4]. The data are strongly bimodal. They correspond to short and long eruption regimes, and we analyse them on the original scale in minutes. We run two chains from setting-adapted starting values. One start swaps the labels and uses the 20% and 80% quantiles of the durations. The other spreads the means beyond the observed range. Each chain runs for 20,000 iterations with 5,000 burn-in.

The posterior places the two components at 2.05 and 4.30 minutes. The proportion of short eruptions is 0.36, and the residual standard deviation is 0.37 minutes. This agrees well with the known two-regime structure of the geyser. On the original scale the components are well separated. The chains never switch labels (0 switches), and the minimum bulk ESS is 27,119. The posterior summaries are reliable.

Table 9 adds convergence diagnostics to the common-variance summaries. The Faithful panel in Figure 3 has only 1 of 272 durations with a lower-component probability between 0.2 and 0.8. This supports clear membership under the fitted model. The eruption durations are from the R `datasets` package [4].

**Table 9.** Faithful: common-variance summaries and diagnostics from both ordered chains. MCSE is for the posterior mean.

| Parameter | Median | 95% interval | Rhat | Bulk ESS | Tail ESS | MCSE |
|---|---|---|---|---|---|---|
| mu (short) | 2.048 | [1.972, 2.125] | 1.0000 | 27885 | 28254 | 0.0002 |
| mu (long) | 4.297 | [4.241, 4.354] | 1.0001 | 28807 | 28923 | 0.0002 |
| pi (short) | 0.361 | [0.306, 0.419] | 1.0000 | 29092 | 30049 | 0.0002 |
| sigma2 | 0.139 | [0.118, 0.166] | 1.0000 | 27119 | 28882 | 0.0001 |

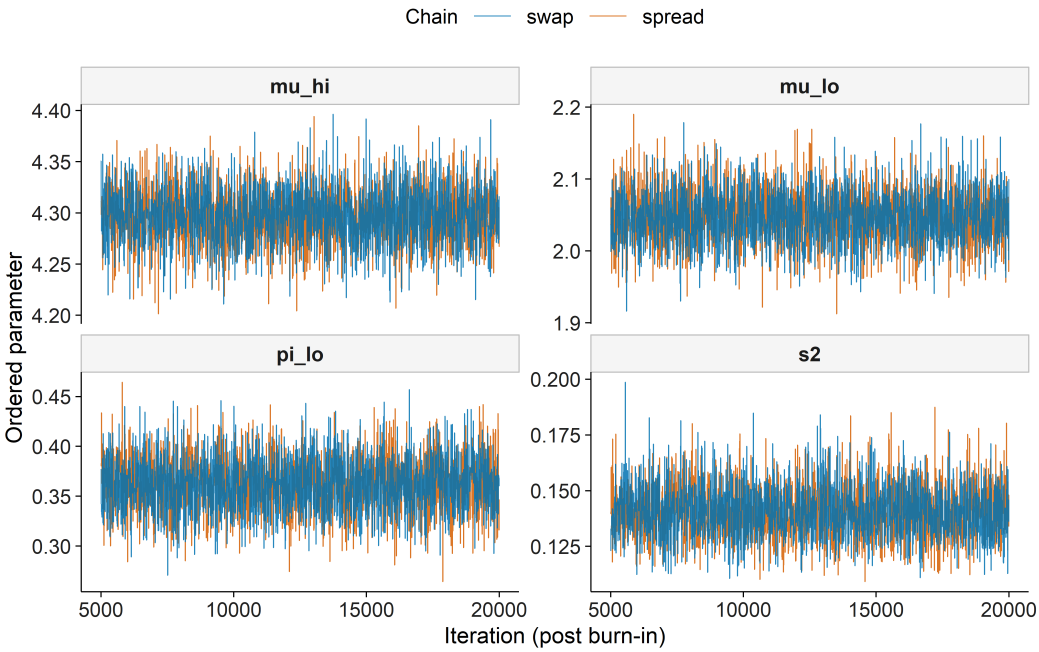



**Figure 6.** Faithful common-variance ordered traces. The two starts give stable, closely agreeing trajectories.

### Posterior predictive check and heteroscedastic extension

Visual inspection of the fitted density indicates a localized discrepancy: the observed short-duration peak is sharper and taller than the fitted lower component, which is broader and lower in that region. The common-variance assumption $\sigma_1^2 = \sigma_2^2$ forces a compromise between tightly clustered short eruptions and more dispersed long eruptions.

To evaluate this quantitatively, we conduct a posterior predictive check [5]. Using regime-specific dispersion statistics ($T_{\text{ratio}} = \text{SD}_{\text{long}}/\text{SD}_{\text{short}}$), the observed spread ratio is 1.540 ($\text{SD}_{\text{short}} = 0.267$, $\text{SD}_{\text{long}} = 0.411$). Across replicated datasets generated under the shared-variance model, the posterior predictive $p$-value is $p_B < 0.001$, confirming the inadequacy of the homoscedastic assumption.

Extending the model to allow component-specific variances ($\sigma_1^2 \neq \sigma_2^2$) via independent Inverse-Gamma updates yields $\sigma_{\text{short}} \approx 0.290$ minutes and $\sigma_{\text{long}} \approx 0.431$ minutes. This heteroscedastic formulation captures the sharp short-eruption peak while avoiding likelihood degeneracy.

### Additional checks for the current saved results

For the predictive check, the observed and each of the 2000 replicated datasets are split at 3 minutes. The two SDs describe duration groups, rather than the latent component SDs. The current run has 0/2000 replicated ratios at least as large as the observed ratio. This supports the spread discrepancy described above; the zero count is a finite-simulation estimate, not proof of a zero tail probability.

The following code defines the SD ratio and calculates how often the replicated ratio is at least as large as the observed one.

```
split_val <- 3
calc_stats <- function(y_vec) {
  s_short <- sd(y_vec[y_vec < split_val])
  s_long <- sd(y_vec[y_vec >= split_val])
  c(sd_short = s_short, sd_long = s_long,
    sd_ratio = s_long / s_short)
}
tail_count <- sum(rep_df$sd_ratio >= obs_stats["sd_ratio"])
p_B <- tail_count / nrow(rep_df)
```

**Table 10.** Common-variance posterior predictive checks. Tail counts give replications at least as large as the observed statistic.

| Statistic | Observed | Rep. median | Rep. 95% interval | Tail count |
|---|---|---|---|---|
| SD short (< 3 min) | 0.267 | 0.366 | [0.312, 0.423] | 2000/2000 |
| SD long (>= 3 min) | 0.411 | 0.377 | [0.325, 0.435] | 238/2000 |
| Ratio (SD long / SD short) | 1.540 | 1.035 | [0.867, 1.231] | 0/2000 |

The original discussion quotes SDs of 0.295 and 0.412 minutes. The current saved output gives posterior SD medians of 0.290 and 0.431 minutes. The following tables and density overlay use the current saved output. Allowing different variances does not itself remove likelihood degeneracy; the proper inverse-gamma priors penalise variances approaching zero.

The unequal-variance model keeps the same mean and weight priors and uses independent InvGamma$(2, 1)$ variance priors. Allocations and mean updates use each component's variance. When ordering the means, each variance moves with its component. Its variance update is

$$\sigma_k^2 \mid - \sim \text{InvGamma}\left(2 + \frac{n_k}{2},\ 1 + \frac{1}{2}\sum_{i:Z_i=k}(y_i - \mu_k)^2\right), \qquad k = 1, 2.$$

Each component variance is updated using its own observation count and residual sum of squares.

```
s2_1_c <- 1 / rgamma(1, shape = alpha0 + n1 / 2, rate = beta0 + sse1 / 2)
s2_2_c <- 1 / rgamma(1, shape = alpha0 + n2 / 2, rate = beta0 + sse2 / 2)
```

**Table 11.** Faithful unequal-variance posterior summaries from both ordered chains.

| Parameter | Median | 95% interval |
|---|---|---|
| mu (short) | 2.031 | [1.970, 2.094] |
| mu (long) | 4.285 | [4.217, 4.350] |
| pi (short) | 0.354 | [0.299, 0.413] |
| sigma2 (short) | 0.084 | [0.061, 0.117] |
| sigma2 (long) | 0.185 | [0.148, 0.236] |

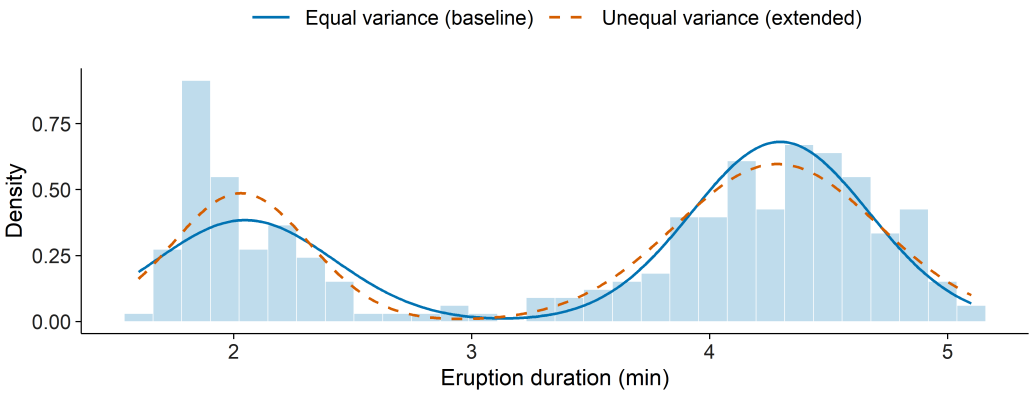



**Figure 7.** Observed Faithful durations with posterior predictive densities under equal and unequal component variances.

The two fits place the means close together. Allowing different variances sharpens the short-duration peak and broadens the long-duration component.

**Table 12.** Posterior medians under the two variance specifications. Means and SDs are in minutes; the weight refers to the short-duration component.

| Parameter | Equal variance | Unequal variance |
|---|---|---|
| Mean (short) | 2.048 | 2.031 |
| Mean (long) | 4.297 | 4.285 |
| Weight (short) | 0.361 | 0.354 |
| SD (short) | 0.373 | 0.290 |
| SD (long) | 0.373 | 0.431 |

Separate convergence diagnostics and predictive replications for the extension are not included in the current script. The reported predictive check evaluates the common-variance model. The overlay shows a more flexible fit, but does not by itself establish model adequacy or verify physical eruption mechanisms.

### Conclusions

The derived Gibbs sampler gives interpretable results under separation. With overlap, a plausible predictive density coexists with uncertain component parameters. Sorting supplies consistent names, but lightly occupied components remain sensitive to the broad mean prior. The occupancy-conditioned calculation illustrates this limitation while leaving the full posterior as the primary result.

For Faithful, the common-variance predictive check detects a spread discrepancy. Allowing different variances changes the two component spreads and gives a more flexible fitted density. That is useful descriptive evidence, but the current script does not separately check the extension's convergence or its predictive replications. Neither fit establishes physically distinct populations or verified eruption mechanisms.

**Limits of the evidence.** One sample per setting cannot establish repeated-sampling bias or coverage; two starts cannot guarantee exploration of every mode. Sorting does not cure weak identification, and occupancy conditioning changes the target. The 3-minute duration split is a chosen model-checking statistic. The extension changes the variance-prior structure, and the density overlay alone does not establish overall model adequacy.

**Reproducibility.** `R/task2.R` regenerates the chains, CSVs, PNGs and `results/task2_sim.RData`. The report reads saved results without sampling. The common-variance chain objects record starts, seeds and priors. Their raw and ordered diagnostics, the common-variance predictive replications, the unequal-variance draws and session information are included. Display thinning affects plots only; primary summaries use all retained draws.

## References


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[2] M. Stephens, "Dealing with label switching in mixture models," *Journal of the Royal Statistical Society: Series B*, vol. 62, no. 4, pp. 795–809, 2000, doi: 10.1111/1467-9868.00265.

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